A MORPHISM CARRYING A STRICTLY POSITIVE WEIGHTING IS UNIVERSALLY OPEN

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 22318-22334, immediately following the statement at 22311-22316.

Editorial note. The proof cites one chapter lemma and one chapter section by \ref; both are navigation pointers into the same chapter.

Editorial note. The proof is short but every reduction is displayed; no step is deferred to the author's reader.

Allowed standard prerequisites (editorial).

- Invariance of weightings under base change. chapter lemma lemma-weighting-base-change, cited at proof line 22320. Reduces universal openness to openness of the morphism itself.
- Etale localization of quasi-finite morphisms. chapter Section section-etale-localization, cited at proof line 22327. After passing to an etale neighbourhood of a point of the target, makes the morphism finite over an open neighbourhood of a chosen point.
- Definition of a weighting and of its integral. chapter definition definition-weighting, used at proof line 22331. Supplies local constancy of the integral along the finite piece, which is what converts positivity of the weighting into openness of the image.

Source reference. Original lemma 0F3B: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 22173–22173. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original section 04HF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 11898–11898. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to Y$ be a locally quasi-finite morphism. Let $w : X \to \mathbf{Z}$ be a weighting. If $w(x) > 0$ for all $x \in X$, then f is universally open.*

Proof. Since the property is preserved by base change, see Lemma 0F3B, it suffices to prove that f is open. Since we may also replace X by any open of X, it suffices to prove that $f(X)$ is open. Let $y \in f(X)$. Choose $x \in X$ with $f(x) = y$. It suffices to prove that $f(X)$ contains an open neighbourhood of y and it suffices to do so after replacing Y by an étale neighbourhood of y. By étale localization of quasi-finite morphisms, see Section 04HF, we may assume there is an open neighbourhood $U \subset X$ of x such that $\pi = f|_U : U \to Y$ is finite. Then $\int_\pi w|_U$ is locally constant

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and has positive value at y. Hence $\pi(U)$ contains an open neighbourhood of y and the proof is complete. $\square$